

- **1001-auth and 1002-auth (auth sections)** provide username-password authentication for each endpoint.
- **1001 and 1002 (AOR sections)** manage how each endpoint registers and specifies the maximum number of active registrations allowed.

After adding the configuration code to the *pjsip.conf*, save the file. Now open the Asterisk console using:

```
asterisk -rvvvv
```

Type the commands *pjsip reload* and *pjsip show aors* to check the address of the records created. Refer to Figure 1 to see the output.

We will also need a SIP softphone or hardphone to register this account. Softphones like Ekiga, Zoiper or x-lite can work. Hardphones like Grandstream, DINSTAR, Cisco, etc, can also be used

We are going to use the Zoiper softphone as an example in our setup. The username and password will be the same as we defined in the *pjsip.conf*. Domain/outbound IP will be the server IP where Asterix is running. In my case the IP is 192.168.1.73.

Once you have successfully connected your softphone to the server, you can check your connection on it by running the following on the Asterisk console:

```
pjsip show aors
```

Refer Figure 4 for the output.

Once we have done this, it's time to make a dial plan. A dial plan is essentially a script or set of instructions that defines how calls are handled within the Asterisk system. It's one of the core components of Asterisk and dictates how incoming and outgoing calls are processed. Configuration in a dial plan defines instructions like:

- Directing calls based on time of the day
- Routing calls based on caller ID
- Connecting to external systems, such as CRM or ticketing systems and many more such use cases

We will be creating our dial plan configuration in the */etc/asterisk/extensions.conf* file. Open the file in a text editor, add the following code, and save the file:

```
exten => _10XX,1,Answer()
same => n,MixMonitor(/home/internal_
call/${STRFTIME(${EPOCH},,%Y-%m-%d)}/${EXTEN}_${STRFTIME(${EPOCH},,%Y-%m-%d_%H-%M-%S)}_${EXTEN}_${UNIQUEID}.wav
)
same => n,dial(pjsip/${EXTEN})
same => n,Hangup()
```

Next, we need to restart Asterisk to reload the new dial plan:

```
asterisk -rx "dialplan reload"
```

```
root@test-VirtualBox: ~
Aor: <Aor.....> <MaxContact>
Contact: <Aor/ContactUri.....> <Hash.....> <Status> <RTT(ms)..>
=====
Aor: 1001 1
Aor: 1002 1
Objects found: 2
-- Added contact 'sip:1001@192.168.1.40:59207;transport=UDP;rinstance=387fb09ae14e710a' to AOR '1001' with expiration of 60 seconds
== Endpoint 1001 is now Reachable
test-VirtualBox*CLI> pjsip show aors
Aor: <Aor.....> <MaxContact>
Contact: <Aor/ContactUri.....> <Hash.....> <Status> <RTT(ms)..>
=====
Aor: 1001 1
Contact: 1001/sip:1001@192.168.1.40:59207;transport=U 59ddeff648 NonQual nan
Aor: 1002 1
Objects found: 2
test-VirtualBox*CLI>
```

Figure 4: Softphone connection

■■ Continued to page...98